

Структура таблиц GP

Цель

Научиться создавать таблицы с разными типами хранилищ и сжатием.

Типы данных

Name	Alias	Size	Range	Description
bigint	int8	8 bytes	-9223372036854775808 to 9223372036854775807	large range integer
bigserial	serial8	8 bytes	1 to 9223372036854775807	large autoincrementing integer
bit [(n)]		<i>n</i> bits	bit string constant	fixed-length bit string
bit varying [(n)] ¹	varbit	actual number of bits	bit string constant	variable-length bit string
boolean	bool	1 byte	true/false, t/f, yes/no, y/n, 1/0	logical boolean (true/false)
box		32 bytes	((x1,y1),(x2,y2))	rectangular box in the plane - not allowed in distribution key columns.
bytea ¹		1 byte + <i>binary string</i>	sequence of octets	variable-length binary string
character [(n)] ¹	char [(n)]	1 byte + <i>n</i>	strings up to <i>n</i> characters in length	fixed-length, blank padded

Типы данных

circle		24 bytes	<(x,y),r> (center and radius)	circle in the plane - not allowed in distribution key columns.
date		4 bytes	4713 BC - 294,277 AD	calendar date (year, month, day)
decimal [(p, s)] ¹	numeric [(p, s)]	variable	no limit	user-specified precision, exact
double precision	float8	8 bytes	15 decimal digits precision	variable-precision, inexact
	float			
inet		12 or 24 bytes		IPv4 and IPv6 hosts and networks
integer	int, int4	4 bytes	-2147483648 to +2147483647	usual choice for integer
interval [fields] [(p)]		16 bytes	-1780000000 years to 1780000000 years	time span
json		1 byte +	json of any length	variable unlimited length

Типы данных

Name	Storage Size	Description	Low Value	High Value	Resolution
timestamp [(p)] [without time zone]	8 bytes	both date and time (no time zone)	4713 BC	294276 AD	1 microsecond / 14 digits
timestamp [(p)] with time zone	8 bytes	both date and time, with time zone	4713 BC	294276 AD	1 microsecond / 14 digits
date	4 bytes	date (no time of day)	4713 BC	5874897 AD	1 day
time [(p)] [without time zone]	8 bytes	time of day (no date)	00:00:00	24:00:00	1 microsecond / 14 digits
time [(p)] with time zone	12 bytes	times of day only, with time zone	00:00:00+1459	24:00:00-1459	1 microsecond / 14 digits

Создание таблицы с разными типами

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
);
```

Добавление оптимизации вставки

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
) WITH (APPENDOPTIMIZED = true);
```

Добавление оптимизации вставки

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
    ) WITH (APPENDOPTIMIZED = true);
```

Добавление различной ориентации

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
) WITH (APPENDOPTIMIZED = true, ORIENTATION = 'row'/'column');
```

Добавление сжатия

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
) WITH (APPENDOPTIMIZED = true, ORIENTATION = 'row'/'column', COMPRESSTYPE = 'zlib', COMPRESSLEVEL = 5);
```

Добавление сжатия

```
CREATE TABLE all_datatypes_example (  
    id serial,  
    data_char char(10),  
    data_varchar varchar(50),  
    data_text text,  
    data_int integer,  
    data_smallint smallint,  
    data_bigint bigint,  
    data_decimal decimal(10, 5),  
    data_numeric numeric(15, 7),  
    data_real real,  
    data_double double precision,  
    data_boolean boolean,  
    data_date date,  
    data_time time without time zone,  
    data_timestamp timestamp without time zone,  
    data_timestamptz timestamp with time zone,  
    data_bytea bytea,  
    data_json json,  
    data_xml xml  
) WITH (APPENDOPTIMIZED = true, ORIENTATION = 'row'/'column', COMPRESSTYPE = 'quicklz');
```

Итог

Научились создавать таблицы с разными типами хранилищ и сжатием.